Definition. *In a connected graph* G*, the* ***distance*** *between vertices* u, v *is*

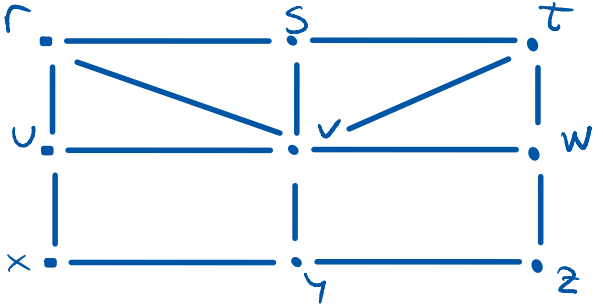
$$d(u,v) := \text{ smallest length of any } u\text{–}v \text{ path in } G.$$

Any two vertices are connected with a path

- A u–v path of length $d(u,v)$ is called a **geodesic**. → u-v path of smallest length.
- The **diameter** of G, denoted $\operatorname{diam}(G)$, is the greatest distance between any two vertices.

Note 1: If G is connected, there are vertices u, v such that $d(u,v) = \operatorname{diam}(G)$.

Example (Exercise 1.12)**.** *For the depicted graph* G*, give an example of each of the following or explain why no such example exists.*



1. *An* x–y *walk of length* 6. xursrvy
2. *A* v–w *trail that is not a* v–w *path.* vsrvw
3. *An* r–z *path of length* 2. No
4. *An* x–z *path of length* 3. No
5. *An* x–t *path of length* $d(x,t)$. xuvt (length 3)
6. *A geodesic whose length is* $\operatorname{diam}(G) = 3$

Take x–t geodesic xuvt

Notation:

- If G is a graph and $v \in V(G)$, then $G - v$ is the graph obtained by removing v along with any incident edges.
- If G is a graph and $e \in E(G)$, then $G - e$ is the graph obtained by removing e from G (keep all vertices).
- If G is a graph and $e \notin E(G)$, then $G + e$ is the graph obtained by adding the edge e to G .

Theorem (Theorems 1.8, 1.9). *Let G be a graph in at least 3 vertices. Then G is connected if and only if it contains two distinct vertices u and v such that $G - u$ and $G - v$ are connected.*

($\Leftarrow$) Assume $u \neq v$ such that $G-u$ and $G-v$ are connected.

WTS G is connected. Take x, y in G. WTS There is an x-y path in G.

Case 1 $\{u,v\} \neq \{x,y\}$. Wlog $u \notin \{x,y\}$. Now x, y are vertices in $G-u$

So there is an x-y path in $G-u$, which is also a path in G.

Case 2 $\{u,v\} = \{x,y\}$. Wlog $u = x$, $v = y$. Want a u-v path in G.

Let $w \neq u, v$ in $V(G)$. Take P' a u-w path in $G-v \subseteq G$

P'' a w-v path in $G-u \subseteq G$

Take $P = P'$ followed by P'' is a u-v walk in G

Now we know, There $\exists$ a u-v path in G ∎

Notation:

- If G is a graph and $v \in V(G)$, then $G - v$ is the graph obtained by removing v along with any incident edges.
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Theorem (Theorems 1.8, 1.9)**.** *Let G be a graph in at least 3 vertices. Then G is connected if and only if it contains two distinct vertices u and v such that $G - u$ and $G - v$ are connected.*

$\Rightarrow$) Exercise : Sketch: 1) Take u, v such that $\operatorname{diam} G = d(u,v)$.

2) Assume $G-v$ disconnected, take x, y in $G-v$ not connected.

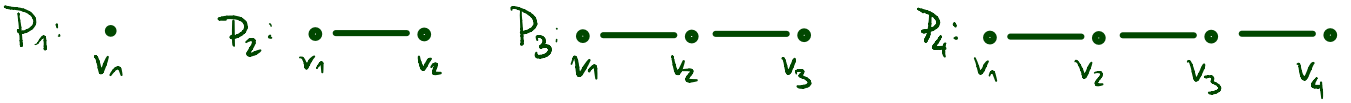
3) Take P' x-u geodesic in G, P'' u-y geodesic in G.

4) Then $v \notin P'$ (otherwise $d(u,x) > d(u,v) = \operatorname{diam} G$) $\Big($ $P' \subseteq G-v$
$v \notin P''$ / $P'' \subseteq G-v$

5) Take P' followed by P'' in $G-v$ to contradict
x and y are not connected in $G-v$.

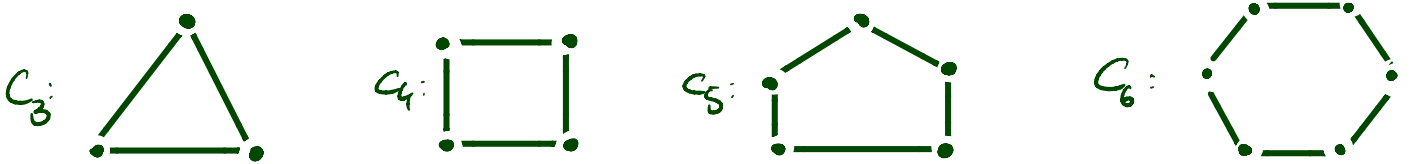
Definition. *For $n \geqslant 1$, the **path** P_n is the graph with vertices $v_1, v_2, \ldots, v_n$ and edges $v_i v_{i+1}$ for $i \in \{1, \ldots, n-1\}$.*

Note that size$(P_n) = n - 1$. $= |E(P_n)|$



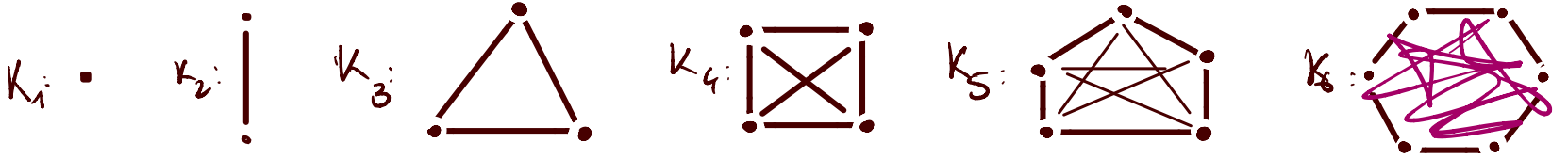
Definition. *For $n \geqslant 3$, the **cycle** C_n is the graph obtained from P_n by adding the edge $v_n v_1$.* $C_n = P_n + v_n v_1$

Note that size$(C_n) = n$. An {**even, odd**} **cycle** is one of {even, odd} size.



Definition. *For $n \geqslant 1$ the **complete graph** (or **clique**) K_n is the graph on n-vertices, where every two vertices are adjacent.*

Note that size$(K_n) = \binom{n}{2} = \frac{n(n-1)}{2}$

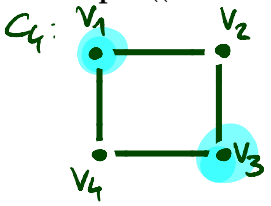


Next: graphs whose vertices can be partitioned in special ways. As a warm up:

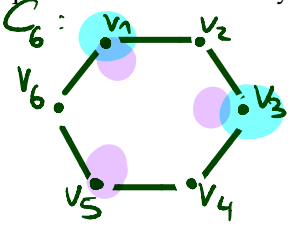
Definition. *An **independent set** on a graph G is a subset X of $V(G)$ such that for any x, y in X, xy is **not** in $E(G)$.*

- **Alternative defn:** the induced subgraph $G[X]$ has no edges ($E(G[X]) = \emptyset$)
- An independent subset is **maximal** if it is **not** properly contained in another independent subset.

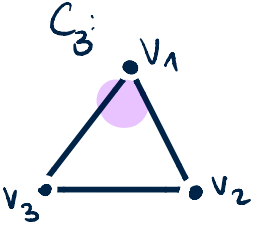
Example ((Maximal) Independent subsets of cycles).



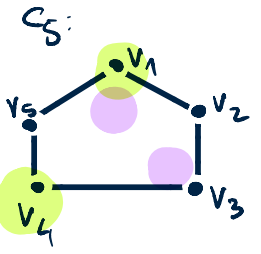
$X = \{v_1, v_3\}$ Maximal size $1 = 2/2$



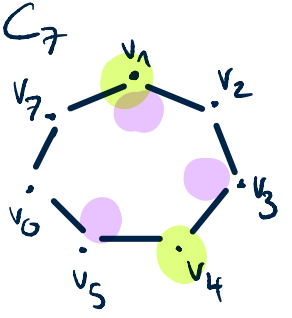
$X = \{v_1, v_3\} \longrightarrow Y = \{v_1, v_3, v_5\}$ $3 = 6/2$



$\{v_1\}$ $1 = \frac{3-1}{2}$



$X = \{v_1, v_4\}$ $Y = \{v_1, v_3\}$ $\frac{5-1}{2}$



$X = \{v_1, v_4\}$ $Y = \{v_1, v_3, v_5\}$ $\frac{7-1}{2}$

Proposition. *For n odd, the largest size of an independent subset for the cycle C_n is $\frac{n-1}{2}$ (achieved e.g. by $\{v_i : i \text{ is odd}, i \neq n\}$).*

For n even, it's $\frac{n}{2}$ → Homework!

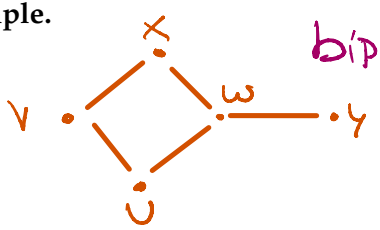
1) $V(G) = X \cup Y$
2) $X \cap Y = \emptyset$

Definition. *A graph* G *is* ***bipartite*** *if there is a partition* $V(G) = X \sqcup Y$ *where* X *and* Y *are independent subsets.*

HW!

- If $V(G) = X \sqcup Y$ is a bipartition and $H \subseteq G$ is a subgraph, then $V(H) = (V(H) \cap X) \sqcup (V(H) \cap Y)$ is a bipartition of H.
- A graph is bipartite if and only if each of its components is bipartite.

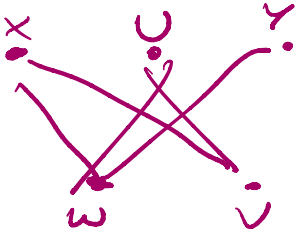
Example.



bipartite?

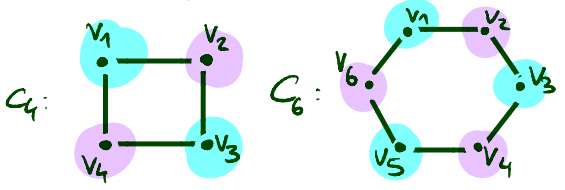
$X =$

$Y =$

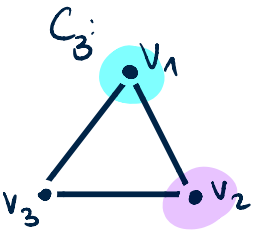


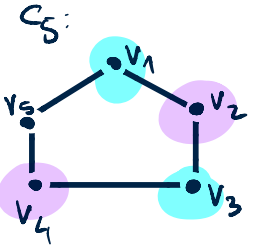
Yes!

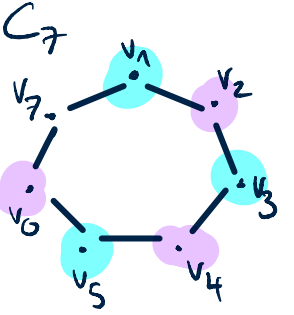
Example (Cycles).



Bipartite!







NOT bipartite!

Lemma. *No odd cycle is bipartite. All even cycles are bipartite.*

Theorem (Theorem 1.12). *A nontrivial graph* G *is bipartite if and only if* G *contains no odd cycles.*

$\Rightarrow$) If G is bipartite, so are all subgraphs, so it contains no odd cycles.

$\Leftarrow$) Wlog G is connected. Assume it contains no odd cycles.

Take $v \in V(G)$. $X = \{v \in V(G) : d(u,v) \text{ is even}\}$, $Y = \{v \in V(G) : d(u,v) \text{ is odd}\}$

WTS $V(G) = X \cup Y$ is a bipartition, Let's see X is independent.

Otherwise, we have v, w in X such that $vw \in E(G)$.

v in $X \Rightarrow d(u,v) = 2k$, take a u-v geodesic $P' = (u = v_0, v_1, \ldots, v_{2k} = v)$

w in $X \Rightarrow d(u,w) = 2\ell$, u-w geodesic $P'' = (u = w_0, w_1, \ldots, w_{2\ell} = w)$



Let x be the last vertex that's in P' and P''

$x \in P' = (v_0, v_1, \ldots, v_{2k}) \rightsquigarrow x = v_i$ for some i

$x \in P'' = (w_0, \ldots, w_{2\ell}) \rightsquigarrow x = w_j$ for some j

Claim: $i = j$

Using the geodesic P' $\quad d(v, x) = d(v_0, v_i) = i$

Using the geodesic P'' $\quad d(v, x) = d(w_0, w_j) = j$ ✓

Consider

$C = (v_i, v_{i+1}, \ldots, v_{2k} = v, w = w_{2\ell}, w_{2\ell-1}, \ldots, w_i = x)$

length of $C = 2k - i + 1 + 2\ell - i = 2(k + \ell - i) + 1$ odd

So G contains an odd cycle! Contradiction. ∎